

Anisotropic creep in CMSX-4 in orientations distant from $\langle 001 \rangle$

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Abstract

Constant load creep tests have been performed on the single crystal nickel base superalloy CMSX-4 in crystallographically unstable orientations distant from the $\langle 001 \rangle$ pole of the standard stereographic triangle at 1023 K (750°C) and varying stress levels. Orientations distant from the $\langle 001 \rangle$ – $\langle 111 \rangle$ boundary had high creep rates and relatively low rupture lives, while the orientations closer to it had relatively low initial creep rates and longer rupture lives. Electron back scatter diffraction patterns, cross section shape change measurements on creep tested specimens, and optical microscopy indicate that the operating slip system is of the type $\{111\}\langle 112 \rangle$ in orientations far from the $\langle 001 \rangle$ – $\langle 111 \rangle$ boundary. The results are explained based on the resolved shear stress acting on the operating slip system taking into consideration the rotations and changes in cross section. A theory based on the ease of generation of superlattice intrinsic stacking faults is proposed to account for the observed creep rates in various orientations. A finite element model based on the slip system approach is used to demonstrate the validity of the experimental results. © 2000 Elsevier Science S.A. All rights reserved.

Keywords: Creep; Single crystals; Superalloys; Anisotropy of creep

1. Introduction

In service single crystal nickel based superalloys used for turbine blade applications often experience high temperatures and stresses such that creep resistance is a prime requirement. Generally superalloy blades are cast with the nominal $\langle 001 \rangle$ orientation parallel to their length, which reduces the incidence of thermal fatigue from temperature variations encountered in operation. Under actual operating conditions, however, the blades experience stresses in a variety of orientations owing to such factors as local hot spots, bending moments and torsional stresses across the blade. In addition the complex cooling system of air channels in the blades also contribute to thermal stress generation in various orientations.

Single crystals are inherently anisotropic and nickel base superalloys are known to exhibit differing creep response as a function of loading orientation [1]. The degree of anisotropy is strongly influenced by both the temperature and stress level. However it is well estab-

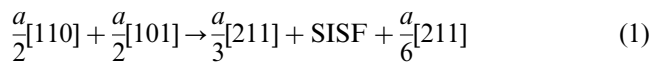
lished that the greatest sensitivity to orientation is observed in the low temperature regime around 1023–1123 K (750–850°C), which is a level typically experienced in the central load carrying webs of many current cooled turbine blades.

Previous work, largely undertaken on first generation superalloys, has established the most common deformation mechanisms observed in the various temperature regimes. For orientations close to $\langle 001 \rangle$, Leverant et al. [2] indicated that in the first generation single crystal superalloy MAR-M200, at 1033 K (760°C) primary creep occurs via a $\frac{2}{3}\langle 112 \rangle\{111\}$ shear of γ' whilst secondary creep was assumed to be on $\frac{2}{3}\langle 110 \rangle\{111\}$ slip systems from activation energy calculations. MacKay and Maier [1] studied the influence of orientation on the stress rupture properties of single crystal MAR-M247 at 1047 K (774°C) and concluded the same mechanisms for primary and secondary creep as Leverant et al. [2]. Significantly they found that stress rupture lives were strongly influenced by the levels of lattice rotation required in the creep specimens to produce conditions favourable for multiple slip. It was suggested that this largely controlled the strain level accumulated during primary creep, which was inversely related to rupture life.

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Caron et al. [3] carried out detailed transmission electron microscopy (TEM) analysis of CMSX-2 containing 7.9% Cr, 4.6% Co, 7.9% W, 0.6% Mo, 6.0% Ta and 5.62% Al, 1.0% Ti as the main alloying elements in Ni. They confirmed the operation of $\frac{a}{3}\langle 112 \rangle\{111\}$ slip in γ' early in creep by the dissociation of $\frac{a}{2}\langle 110 \rangle$ dislocations at γ/γ' interfaces although pointing out that the presence of superlattice intrinsic stacking faults (SISFs) did not necessarily imply this was the principal deformation mechanism, but rather one which left a clearly visible trail. A possible mechanism for generation of the $\frac{a}{3}\langle 112 \rangle$ dislocations arises from the association of dissimilar $\frac{a}{2}\langle 110 \rangle$ dislocations on the same slip plane. Such a reaction was observed by Sass et al. [4] near the $[011]$ orientation. A typical reaction was reported as:



Kuttner and Feller-Kniepmeier [5,6] performed creep tests in the $[011]$ orientation at 1033 K (760°C) and 680 MPa on SRR 99 and observed that initial slip occurs in the matrix on the four highly stressed $\langle 110 \rangle\{111\}$ slip systems. Dislocation activity was confined to matrix channels in which the coherency stresses combined in an additive fashion with those applied externally, leading to high resolved shear stresses. Only in the secondary stages of creep was shearing of the γ' precipitates due to $\frac{a}{3}\langle 112 \rangle\{111\}$ slip reported. At the higher temperature of 1123 K (850°C), Sass et al. [4] have observed cutting of the γ' phase by $\frac{a}{3}\langle 112 \rangle\{111\}$ slip near $\langle 001 \rangle$ and $\langle 011 \rangle$ orientations in the second-generation Re containing single crystal superalloy CMSX-4 during primary creep at high stresses. Matan et al. [7] have reported the simultaneous operation of more than one $\langle 112 \rangle\{111\}$ slip systems in CMSX-4 during the early stages of primary creep at 1023 K (750°C) and 750 MPa close to $\langle 001 \rangle$ orientation.

A set of equations to describe the creep behaviour of isotropic nickel based alloys were developed based on the work by Dyson et al. [8–10], Henderson and McLean [11] and Ion et al. [12]. Ghosh et al. [13] extended this model to account for anisotropic creep in single crystals by a slip system approach. The total

strain was obtained as the sum of the shear components on allowed slip systems. Pan et al. [14] and Henderson et al. [15] modelled the cross section shape changes of specimens creep tested in the temperature range of 1023–1223 K (750–950°C) using this model. In orientations close to $\langle 111 \rangle$ good agreement between the model and the experimental data was only obtained by the activation of cube slip on $\langle 011 \rangle\{100\}$ system, which has the highest Schmid factor of 0.47 in $\langle 111 \rangle$ orientation. Hence they originally proposed that a cube slip type mechanism was active in the vicinity of $\langle 111 \rangle$, while octahedral slip was active in other orientations with a small region of uncertainty along the intersection of these areas.

Previous work on the mechanisms of creep in various stages of creep has been largely based on TEM evidence and is necessarily not in agreement with the macroscopic experimental evidence and modelling. The purpose of this current paper is to further explore the creep behaviour of CMSX-4 under varying stress levels at 1023 K (750°C) in unstable orientations within the standard stereographic triangle, using electron back scatter diffraction patterns (EBSP), shape change measurements and TEM. The results are discussed with respect to current modelling activities aimed at prediction of the creep response.

2. Materials and experimental procedure

2.1. Creep test specimens

CMSX-4 single crystal rods were provided by Rolls-Royce plc. in the fully heat treated condition as finger castings. The heat treatment comprised a three-step solution treatment consisting of 1 h soaking at 1553 K (1280°C), 2 h at 1563 K (1290°C) and 6 h at 1573 K (1300°C). This was followed by a two-step ageing treatment of 6 h at 1413 K (1140°C) and 16 h at 1143 K (870°C). This standard heat treatment produced a γ' volume fraction of 70% in the form of regular cubes with an edge length of approximately 0.45 μm . The orientations of all the rods were determined by the automated Laue backscatter X-ray diffraction tech-

Table 1
Initial orientations and test conditions of various specimens

Specimen	Angle from $\langle 001 \rangle$	Angle from $\langle 001 \rangle$ – $\langle 011 \rangle$	Tests conditions	
			Temperature (°C)	Stress (MPa)
A	34	40	750	750
B	47.9	32	750	720
C	42.2	22.3	750	675 & 750
D	41.3	3.6	750	720
E	37.5	0.4	750	750

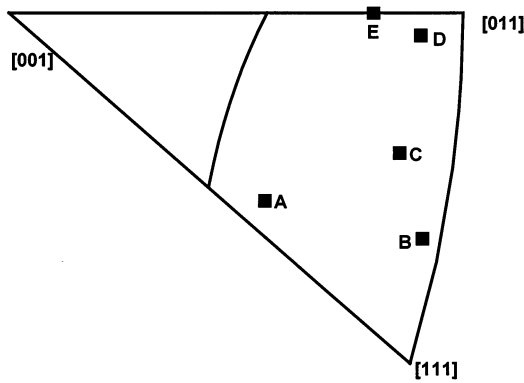


Fig. 1. Orientation of test specimens in the standard stereographic triangle.

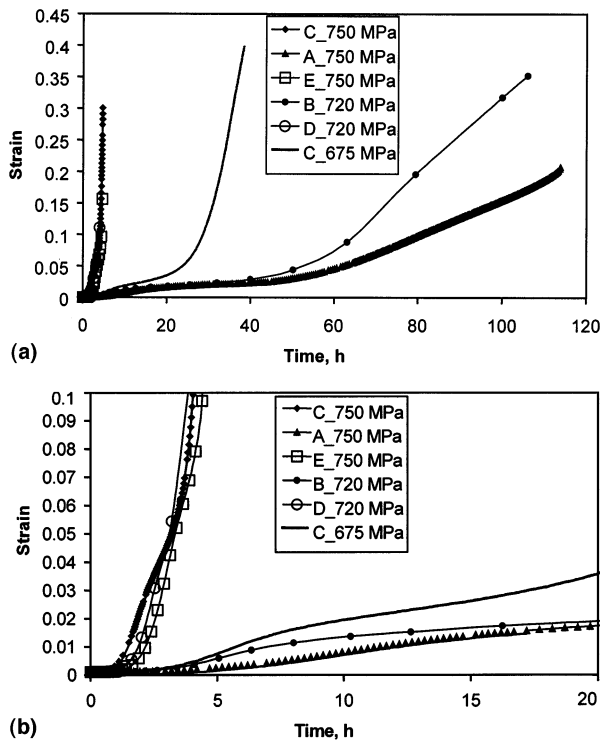


Fig. 2. (a): Creep curves at 1023 K (750°C) for various orientations (A, B, C, D, E) and stresses. (b): Magnified initial region of a.

nique, SCORPIO [16]; these are given in Table 1 and illustrated in Fig. 1 on a standard stereographic triangle. Conventional creep specimens with a gauge length of 28 mm were machined from the bars. Constant load creep tests were performed at 1023 K (750°C) over a range of stress levels. The various test conditions are also listed in Table 1.

2.2. Cross section shape measurements

Following testing the fractured specimens were mounted in a jig such that the cross section at points along the gauge length could be traced using a Leitz PMM12106 computer numerical control machine capa-

ble of measuring the three dimensional coordinates. Measurements were started at a reference point on the gauge length, which was subsequently used to determine the direction of the crystallographic axes on the shape change plot (Appendix A). The specimen was rotated through 360° producing a plot of the surface of the specimen. The procedure was repeated at 2-mm intervals along the gauge length until the fracture surface was reached.

2.3. Lattice rotation measurements

Lattice rotations occurring in the specimens during deformation were determined by EBSD on a JEOL 5800LV scanning electron microscope (SEM). The creep-tested specimens were machined to yield a 4 mm × 3 mm rectangular cross section. These bars were then mounted and mechanically ground using standard metallographic procedures with final polishing using colloidal silica. The procedures for obtaining EBSD in a SEM are described in Refs. [17,18]. The Kikuchi patterns obtained were indexed using the TSL OIM software [19] to find the orientation of the crystal with an accuracy of 1°. Average orientations were determined by scanning a square area approximately 225 μm² at 0.5 mm intervals along the entire gauge length of the specimen.

3. Results

The creep curves for the various tests conducted in different orientations and stresses at 1023 K (750°C) are shown in Fig. 2a. Fig. 2b shows an expanded view of the initial portion of the curves.

It is clear that orientations C, D, E display very high creep rates compared to orientations A and B at 750 and 720 MPa, respectively. The orientations C, and E at 750 MPa, and D at 720 MPa failed in less than 5 h. Both D and E show a monotonic increase in creep rate throughout their lives. C at 750 MPa contained a short period of decreasing strain rate, but this is limited. Orientations B and A produce a more typical creep response with a limited range of primary creep followed by a transition through secondary to an extended regime of tertiary deformation. A further test in orientation C at the lower stress of 675 MPa yielded a curve with a shape similar to that of A and B at the higher stress levels.

Fig. 3 shows the lattice rotation measurements performed on the creep tested samples A, C and E. It can be seen that orientations C and E rotate towards the principal <211> pole. For orientation A, however, this progress is halted by the presence of the [001]–[111] symmetry boundary, which causes rotation to deviate from its original course towards the [111] pole.

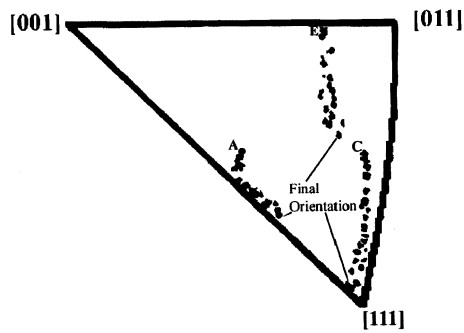


Fig. 3. Measured EBSD lattice rotations for specimens A and C.

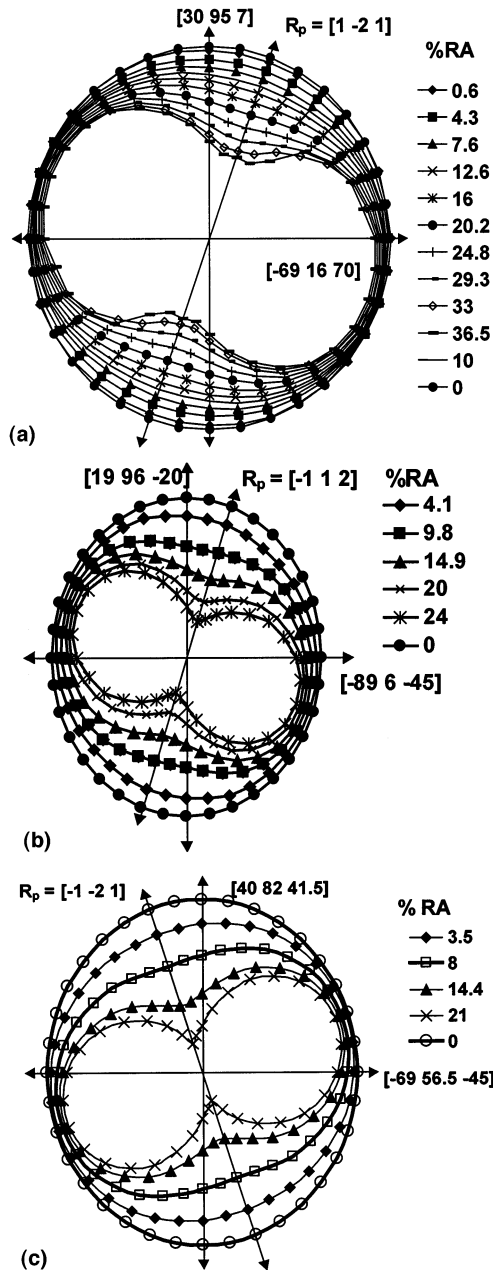


Fig. 4. Specimen cross section shape change at 1023 K (750°C) and 750 MPa for the orientation (a) C, (b) A and (c) E. R_p = Projection of the slip direction into the plane perpendicular to the tensile axis (refer Fig. 5). %RA = percentage reduction in area.

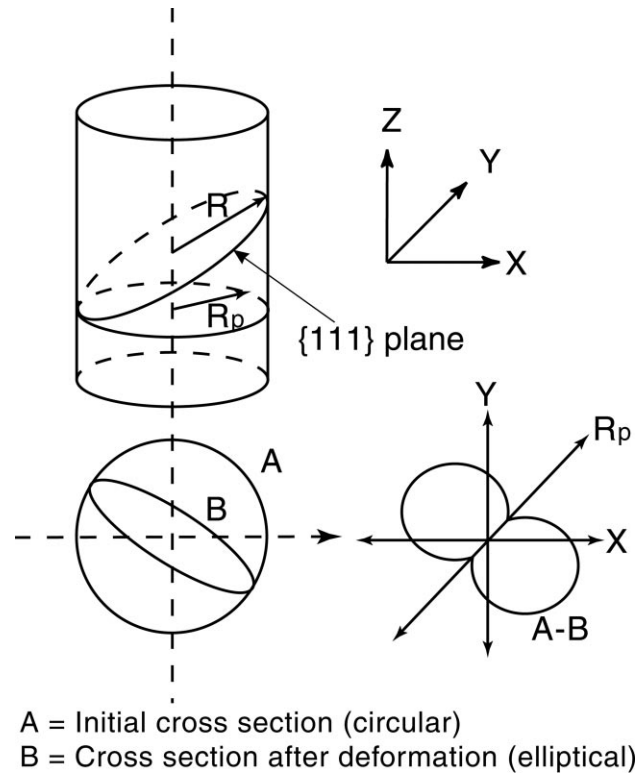


Fig. 5. Illustration of the projection of the slip direction R in a $\{111\}$ plane into a plane perpendicular to the tensile axis (R_p) in a single crystal.

Fig. 4(a–c) illustrates measured specimen shape cross section changes. The cross section shape shown here is not the actual shape of the specimen, but is obtained by subtracting a reference circle of constant diameter from the actual shape. This procedure is used because it amplifies visually the effect of the shape change (see Fig. 5 and Appendix A). For specimens C, and E, far from the $\langle 001 \rangle$ – $\langle 111 \rangle$ boundary, these measurements confirm the observation that rotation is towards the principal $\langle 211 \rangle$ pole with the highest resolved Schmid factor, as the direction in which maximum constriction occurs (R_p in Fig. 5) is the projection of the slip direction (R in Fig. 5) in the plane perpendicular to the tensile axis. For specimen A closer to the $\langle 001 \rangle$ – $\langle 111 \rangle$ boundary, the slip direction is found to coincide with the resultant of the two principal $\langle 110 \rangle$ directions.

Fig. 6 shows a typical fracture surface of the specimens C, D and E, which all failed by a clean shear. The surfaces of all the creep specimens far from the $\langle 001 \rangle$ – $\langle 111 \rangle$ boundary, which had not been badly oxidised at these low temperatures, revealed a single set of slip bands parallel to the shear fracture surface of the specimen as shown in Fig. 7, indicating the operation of a single dominant slip plane. For the specimen with orientation A, closer to the $\langle 001 \rangle$ – $\langle 111 \rangle$ boundary, two slip planes were visible in the failed specimen.

Significantly the fracture surface (Fig. 8) for this orientation also contained shear on these two planes providing further evidence that duplex slip had occurred prior to failure, but that deformation had remained



Fig. 6. Typical clean shear fracture observed in specimens with orientations B, C, D and E.

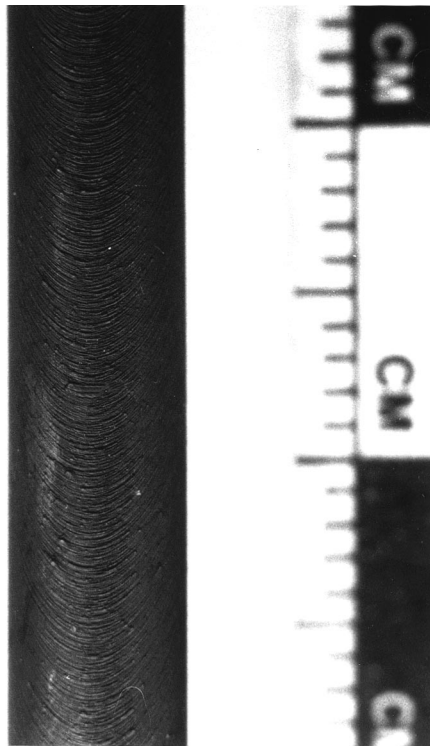


Fig. 7. Parallel slip bands indicative of single slip on specimens tested at 1023 K (750°C) and 750 MPa.

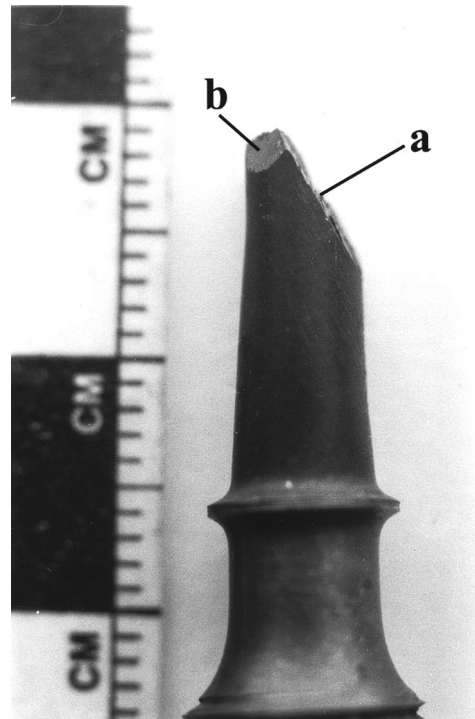


Fig. 8. Fracture surface of A oriented specimens revealing two slip planes a and b.

inhomogeneously associated with the two dominant slip planes.

4. Discussion and microstructural analysis

The EBSD lattice rotation measurements and shape change measurements, Figs. 3 and 4, respectively, show that the orientations C and E tested in this program rotate towards the [211] pole confirming that the resulting operating slip system is of the type $\langle 211 \rangle \{111\}$.

Orientations further away from the $\langle 001 \rangle - \langle 111 \rangle$ boundary and closer to the $\langle 011 \rangle - \langle 111 \rangle$ boundary have higher creep rates as can be seen in the case of the C, D and E orientations tested at the higher stresses. The A and B orientations which are closer to the $\langle 001 \rangle - \langle 111 \rangle$ boundary have very similar initial creep rates (although the applied stress is greater by 30 MPa for A) which are both much lower than that of the other orientations studied here.

The EBSD measurements of Ardakani et al. [20,21] at 1023 K (750°C) and 1123 K (850°C) on a number specimens of SRR 99 and a few of CMSX-4 show rotations towards $\langle 211 \rangle$ pole in similar regions of the standard stereographic triangle. Modelling studies utilising $\langle 110 \rangle \{111\}$ and $\langle 112 \rangle \{111\}$ type of slip systems yielded similar results, which were in agreement with experimental observations. Microscopic observations

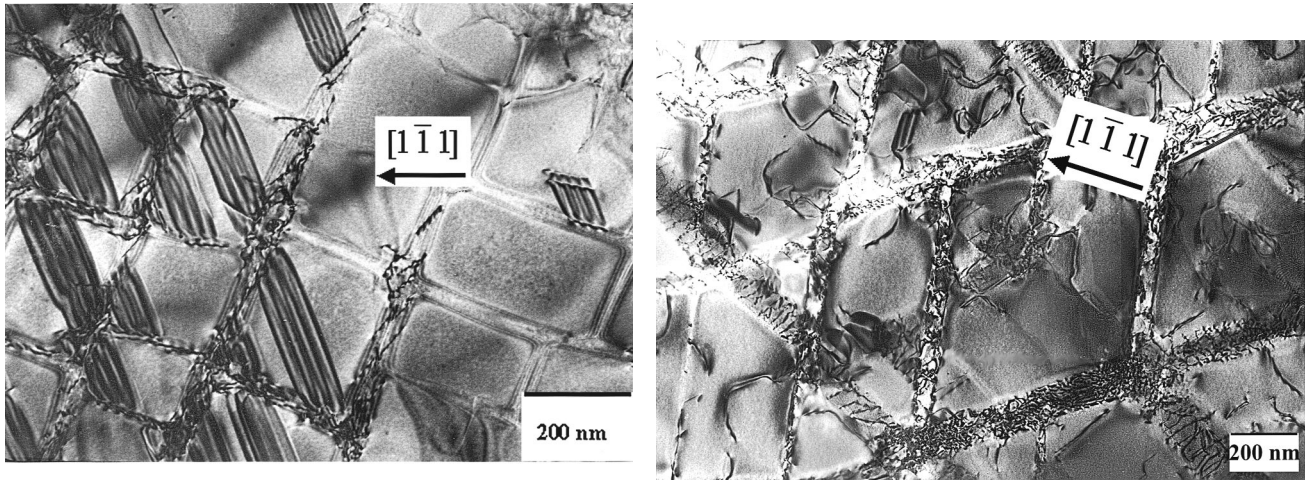


Fig. 9. (a): Typical transmission electron microscope microstructure of specimens, C, D, and E, creep tested at 1023 K (750°C). Beam direction $\sim [1\bar{2}3]$. (b): Typical transmission electron microscope microstructure of specimen A, creep tested at 1023 K (750°C). Beam direction $\sim [\bar{1}12]$.

revealed twinning however, which they indicated could not account for the large strains observed in the crystals. Hence they concluded that deformation was based on $\langle 110 \rangle \{111\}$ type of slip systems.

TEM investigation of the crept specimens C and E a typical example of which is shown in Fig. 9(a), has shown prevailing extended SISFs cutting the γ' particles as a result of the passage of $\frac{2}{3}\langle 112 \rangle$ dislocations, which is consistent with the rotation measurements. Interestingly a foil from specimen A revealed predominantly dislocation climb and glide in the γ channels and anti phase boundary (APB) shearing of γ' , while very few SISFs were observed in γ' Fig. 9(b). No twinning was observed in any of the specimens tested in this programme.

As it is apparent from the experimental evidence that $\langle 112 \rangle \{111\}$ slip controls deformation in the orientations C, D, E, but $\langle 110 \rangle \{111\}$ slip in A, an initial attempt was made to rationalise the different creep rates that are observed by considering the resolved shear stresses (RSS) acting on these systems. Fig. 10 shows the initial orientations superimposed on the Schmid factor contours for the primary $\langle 112 \rangle \{111\}$ slip system. It is clear that high creep rates correspond to locations where the Schmid factors are high although this does not account for changes, which occur during creep and rotation.

Fig. 11a shows the variation of the nominal shear stress with strain for the various orientations. The variation with strain was calculated using the relationship between the angles made by the slip plane and the slip direction, assuming single slip and is given by Eq. (4) as follows:

$$\frac{L_1}{L_0} = \frac{\sin \lambda_0}{\sin \lambda_1} = \frac{\cos \chi_0}{\cos \chi_1} \quad (2)$$

where λ is the angle between glide direction and the tensile axis; χ is the angle between the plane normal and the tensile axis; L_1 is the elongated gauge length; and L_0 is the original gauge length.

$$\text{Initial Schmid factor} = \cos \lambda_0 \cdot \cos \chi_0 \quad (3)$$

From Eqs. (2) and (3)...

$$\begin{aligned} \text{Final Schmid factor} &= \cos \lambda_1 \cdot \cos \chi_1 \\ &= \frac{\cos \chi_0}{(1 + \varepsilon)} \sqrt{1 - \frac{\sin^2 \lambda_0}{(1 + \varepsilon)^2}} \end{aligned} \quad (4)$$

where ε is the nominal strain.

In Fig. 11b the same data is re-plotted with a correction for the increase in stress occurring during the load controlled creep test, the stresses are therefore the true resolved shear stresses. It is pointed out that the discontinuous line in Fig. 11a, b corresponds to the RSS for

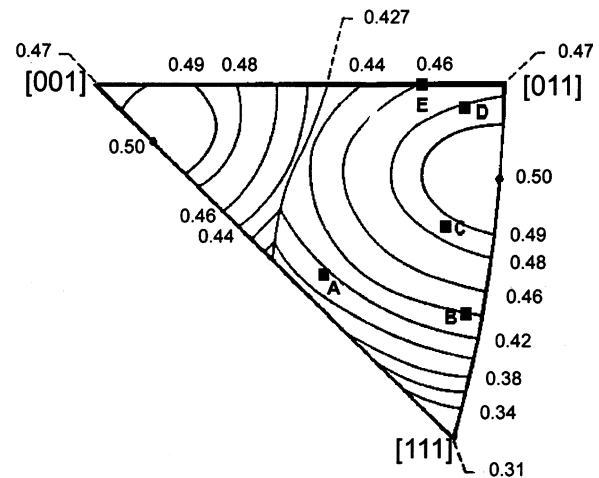


Fig. 10. Initial orientations of specimens superimposed on the Schmid factor contours for $\{111\}\langle 112 \rangle$ slip (modified from [1]).

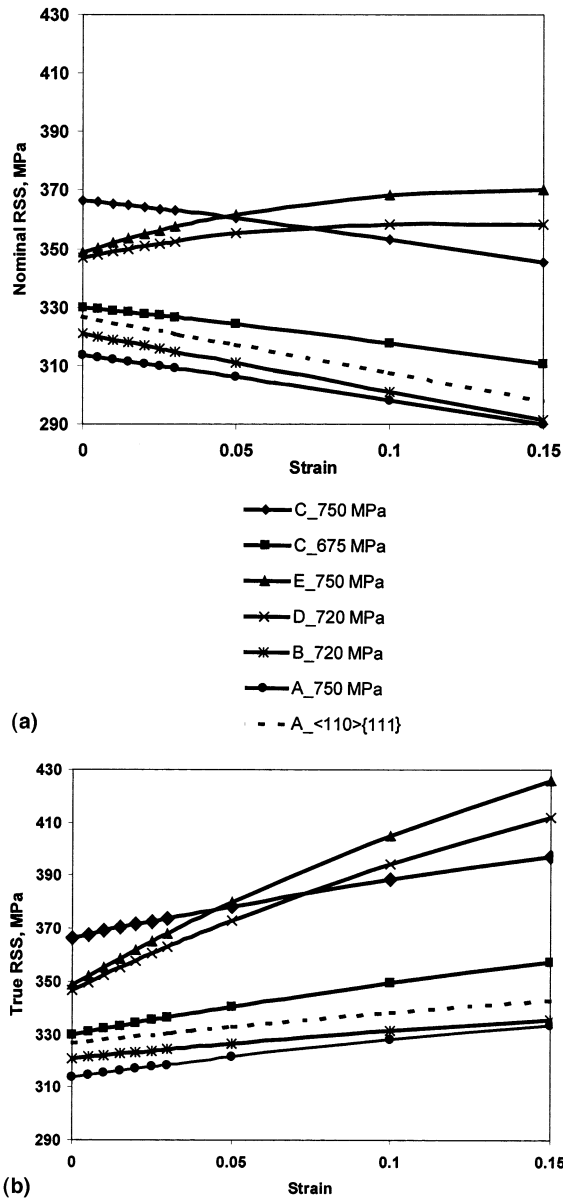


Fig. 11. (a): Variation of nominal resolved shear stress (RSS) on the $\langle 112 \rangle \{111\}$ type slip system with strain for various orientations. (Discontinuous line is the RSS on $\langle 110 \rangle \{111\}$ type of slip system for A). (b): Variation of resolved shear stress on the $\langle 112 \rangle \{111\}$ type slip system with strain for various orientations taking the reduction in area into consideration. (Discontinuous line is the RSS on $\langle 110 \rangle \{111\}$ type of slip system for A).

the orientation A on the $\langle 110 \rangle \{111\}$ type of slip system and is given for the sake of comparison with that on the $\langle 112 \rangle \{111\}$ type of slip system. The RSSs segregate clearly into two regions in a similar fashion to the creep curves in Fig. 2. C and E orientations tested at 750 MPa and the D orientation tested at 720 MPa have higher resolved shear stresses and correspondingly higher creep rates. In the case of orientation C, the initial stress is the highest giving rise to rapid initial creep rates (Fig. 2b). Orientations E and D have lower initial resolved stresses and accordingly reduced initial

deformation rates. During elongation, the stresses all increase due to area reduction, but the effect of the changing Schmid factor is also important. For E and D the Schmid factor increases with rotation whereas for C there is a reduction. This effect manifests itself in the rate at which stress changes (Fig. 11b). After approximately 5% strain the resolved shear stresses for all three orientations are similar. It is thus not surprising to observe similar creep rates at this point. Fig. 12 shows the variation of strain rate with strain in the various orientations tested at 750 and 720 MPa. It is observed that the creep rates segregate into two regions similar to the creep curves, Fig. 2b and the RSS plots, Fig. 11. The creep rates observed are lower by an order of magnitude compared to those obtained by Sass et al. [4], at 1123 K for the similar orientations.

Table 2 lists the various $\langle 112 \rangle \{111\}$ and $\langle 110 \rangle \{111\}$ slip systems with high Schmid factors for the initial orientations A, C and E. In the case of orientation C, there are two $\langle 110 \rangle$ slip directions on the same $\{111\}$ plane with Schmid factors of 0.43 and 0.41, which are very close. Hence they are both activated and can react to form a $[1\bar{2}1](111)$ system which can shear the γ' in a manner similar to that observed by Sass et al. [4]. This is confirmed by the shape change measurement (Fig.

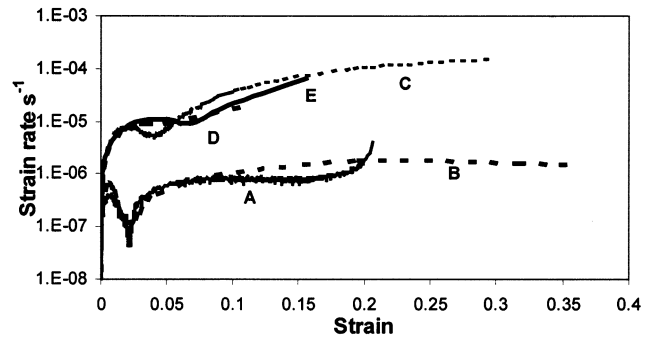


Fig. 12. Variation of creep strain rates with strain in various orientations.

Table 2

Highest Schmid factors for various orientations on $\langle 110 \rangle \{111\}$ and $\langle 112 \rangle \{111\}$ type of slip systems

Initial orientation	Slip system	Schmid factor
A	$[101](\bar{1}11)$	0.43
A	$[011](\bar{1}\bar{1}1)$	0.39
A	$[211](\bar{1}11)$	0.42
C	$[0\bar{1}1](111)$	0.43
C	$[1\bar{1}0](111)$	0.41
C	$[1\bar{2}1](111)$	0.49
E	$[101](111)$	0.46
E	$[1\bar{1}0](111)$	0.34
E	$[110](\bar{1}11)$	0.34
E	$[101](\bar{1}11)$	0.46
E	$[2\bar{1}1](111)$	0.46
E	$[211](\bar{1}11)$	0.46

4(a) and the Appendix A). The resulting $[\bar{1}\bar{2}1](111)$ slip system has a higher Schmid factor (0.49). The dislocation reaction in this case is of the form:



In the case of the specimen E it can be seen from Table 2 that there are two pairs of $\langle 110 \rangle\{111\}$ slip systems, each pair having a common $\{111\}$ plane. Each pair can combine to give $\langle 112 \rangle\{111\}$ resultants again lying in the same plane. In practice only one dominant $\langle 112 \rangle\{111\}$ was observed to be active under TEM. This is most likely a consequence of the fact that once slip occurs, rotation away from the boundary proceeds and only one of the two possible slip systems then becomes dominant. This is confirmed by the shape change measurements in Fig. 4(c) where the direction in which the specimen contracts most corresponds to the projection of the principal $\langle 211 \rangle$ direction.

Orientation C tested at 675 MPa, B tested at 720 MPa and A tested at 750 MPa have significantly lower strain rates. The shear stresses calculated for $\langle 112 \rangle\{111\}$ slip, decreasing from orientation C through B to A are reflected in the observed creep rates. The true shear stresses do not diverge rapidly for these orientations although orientation C does increase significantly relative to the other two indicative of its earlier transition to tertiary creep and ultimate failure.

Orientation A displays the lowest creep rates. The resolved stress for $\langle 112 \rangle\{111\}$ slip on the most highly active system indicates that this is as predicted. In addition, however, deformation for this orientation is complicated by the proximity to the $[001]$ – $[111]$ boundary, which invalidates the large strain calculations of the nominal shear stress (which rely on the assumption of slip on a single system). The presence of double slip may well be expected to lead to enhanced hardening and a further reduction in creep rates, but this does not appear to have a strong influence on the creep rates recorded.

Shape change measurements carried out on specimen A are complicated by the operation of a second slip system but indicate that the slip direction is given by the resultant of the two most highly stressed $\langle 110 \rangle$ directions suggesting that this orientation is deforming on the two principal $\langle 110 \rangle\{111\}$ slip systems that have similar Schmid factors. TEM observations also revealed very few stacking faults suggesting that $\frac{a}{3}\langle 112 \rangle$ shear of γ' is not the main mechanism of deformation. Numerous APB dislocation pairs were observed shearing the γ' precipitates indicating that it is the main mode of deformation.

For the initial orientation of A, there are only two $\langle 110 \rangle\{111\}$ slip systems with a reasonably high Schmid factor (Table 2). However, as opposed to the case of specimen C, these $\langle 110 \rangle\{111\}$ slip systems are on dif-

ferent $\{111\}$ planes. There is no second Burgers vector of the $\langle 110 \rangle$ type, on the same slip plane with a high enough Schmid factor that could be easily activated and lead to interaction forming the $\langle 112 \rangle$ Burgers vectors. The combination of the two $\langle 110 \rangle$ Burgers vectors on the two different $\{111\}$ planes with the highest Schmid factors does not result in a $\frac{a}{3}\langle 112 \rangle$ dislocation on a $\{111\}$ plane. Due to low activation of suitable $\langle 110 \rangle$ dislocations, $\frac{a}{3}\langle 112 \rangle$ dislocations are rarely generated and hence very few SISFs are observed (Fig. 9(b)). The difficulty in generating the $\frac{a}{3}\langle 112 \rangle$ dislocations that allow relatively easy γ' shearing compared to dislocation APBs is the reason for the low initial creep rates observed.

Hence in orientations close to the $\langle 011 \rangle$ – $\langle 111 \rangle$ boundary, stacking fault generation by the association of coplanar $\langle 110 \rangle$ dislocations is favoured, leading to $\frac{a}{3}\langle 112 \rangle$ shear of γ' and resulting in heterogeneous slip. This gives rise to high creep rates shown in Fig. 2. In orientations close to the $\langle 001 \rangle$ – $\langle 111 \rangle$ boundary, the non-availability of highly activated coplanar $\langle 110 \rangle\{111\}$ slip systems decreases the probability of their interaction to produce the required $\langle 112 \rangle$ type of dislocations for easy γ' shear. Hence very few stacking faults are seen in the γ' precipitates and the predominant mechanism is that of dislocation climb and glide through the γ channels and shearing of the γ' precipitates by the higher energy APB mechanism. This leads to the low creep rates observed during the primary and secondary creep stages.

5. Deformation modelling

The general conclusion from the experimental observations is that the resolved shear stress on $\{111\}\langle 112 \rangle$ slip systems dominates the deformation rate for all the creep tests undertaken at 750°C and located near the $\langle 011 \rangle$ – $\langle 111 \rangle$ boundary. These RSSs can be used to account for the relative strain rates measured.

A finite element (FE) slip system plasticity model has been developed for the analysis of single crystal creep using a constitutive creep law developed for CMSX-4 and SRR 99 [23]. The slip system model is based on the classical theory of single crystal plasticity developed by Peirce et al. [22]. Details of the FE modelling procedure and constitutive theory have been described elsewhere [23].

Simulations of crystal rotation during single crystal creep are shown in Figs. 13 and 14 for the initial orientations in which creep tests were performed. These were run using two different scenarios; in the first (Fig. 13) it was assumed that creep is dominated by the single most highly stressed slip system and that system was the only one allowed to be active in the FE model. In the second (Fig. 14) multiple slip was activated and all

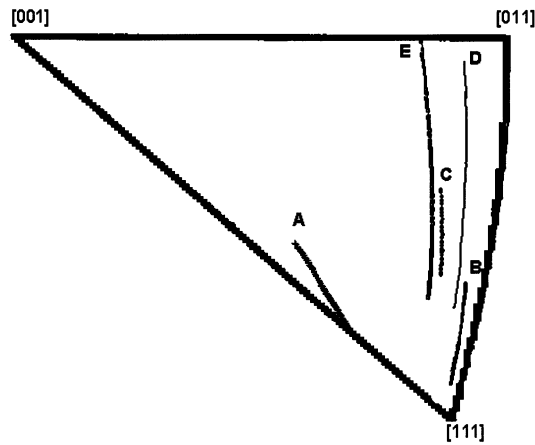


Fig. 13. Simulated lattice rotations using a single $\langle 112 \rangle \{111\}$ slip system.

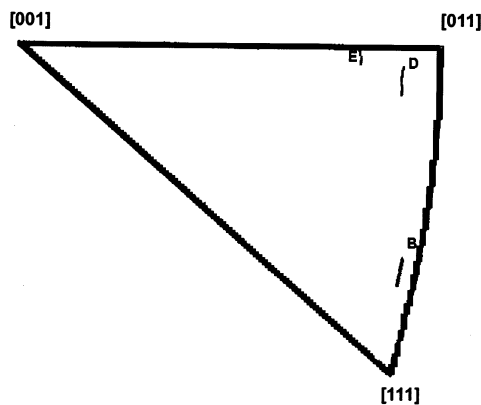


Fig. 14. Simulated lattice rotations using multiple $\langle 112 \rangle \{111\}$ slip system.

slip systems followed the same dependence on resolved shear stress. It should be pointed out here that in these simulations no hardening or softening between the slip systems were included in the model. A comparison of Figs. 13 and 14, demonstrates that the test results are predicted very well by the model when only one slip system is activated whilst when multiple slip is activated the sense of orientation change is usually correct but the magnitude is reduced due to the effect of different slip systems trying to rotate the crystal in different directions. The similarity of single slip predictions and test results suggests that the creep tests are dominated by single slip. From a mechanistic viewpoint the conclusion is that in situations where one slip system experiences a slightly higher resolved shear stress than other systems, it dominates the deformation to a much greater degree than would be expected by a comparison of slip system resolved shear stresses. This situation is caused either by latent hardening by the dominant system on the remaining systems or softening of the dominant slip system. It is not clear as yet whether the fact that off-axis creep is dominated by single slip is due to latent

hardening or self softening. The situation depends on at what stage during creep the significant rotations (a consequence of anisotropic deformation) occur. If they arise during tertiary creep, then they are probably caused by self softening of the dominant slip system and the formation of shear bands. If, however as is likely, single slip starts early in creep it is more probable to be caused by a latent hardening mechanism. The next stage in our FE model development is to include a hardening and/or softening parameter so that the slip and creep strain is confined to the one dominant slip system so that rotations similar to those measured during a creep test are predicted when all the slip systems are active.

The lack of significant $\langle 112 \rangle \{111\}$ slip in orientation A has highlighted the fact that a more complete description of anisotropy at low temperatures must also consider the interaction between $\langle 110 \rangle \{111\}$ slip systems and the ease with which a coplanar $\langle 112 \rangle$ type dislocation can result, enabling γ' shear. This is the subject of ongoing work.

6. Conclusions

At 1023 K (750°C) deformation at stress levels of 675 MPa and above is dominated by $\langle 112 \rangle \{111\}$ slip in orientations close to the $\langle 111 \rangle$ – $\langle 011 \rangle$ boundary. This has been confirmed by rotation and shape change measurements and is supported by macroscopic and TEM observations. The creep rates measured can simply be rationalised with respect to the magnitude of the resolved shear stress on the dominant slip system of the $\langle 112 \rangle \{111\}$.

Easy generation of $\langle 112 \rangle \{111\}$ slip due to the interaction of suitable coplanar $\langle 110 \rangle \{111\}$ pairs of dislocations with high Schmid factors leads to shearing of γ' forming stacking faults and high initial creep rates. On the other hand if the formation of $\langle 112 \rangle \{111\}$ dislocations is difficult due to a low driving force for the generation of suitable $\langle 110 \rangle \{111\}$ dislocations in the γ channels, low initial creep rates are observed.

For orientations near the $\langle 111 \rangle$ – $\langle 011 \rangle$ boundary modelling the deformation by considering strain rates on all the individual $\langle 112 \rangle \{111\}$ slip systems allows accurate prediction of the sense of the orientation change but the magnitude is more accurately predicted by assuming that creep is dominated by the most highly stressed slip system. Further work is in progress to optimise the prediction of the magnitude of the deformation by inclusion of interaction or self softening terms which allow the application of a general model in which all the slip systems are activated, but the most highly stressed dominates. In addition future modelling activity will need to address the interaction of $\{111\} \langle 110 \rangle$ systems in order to account for the ease of $\langle 112 \rangle \{111\}$ slip system generation.

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Appendix A. Verification of the short direction in the shape change plots

Specimen shape change measurements were initiated at a reference point on the gauge length, which was subsequently transferred to the non-deforming grip portion of the specimen.

The orthogonal directions Z , X , and Y in Fig. 5 were determined using EBSP. The short direction SD with respect to the original crystal orientation was subsequently determined by vector addition of X and Y . For the case of specimen C these were found to be

$$Z \sim [66\ 26\ 71]$$

$$X \sim [-69\ 17\ 70]$$

$$Y \sim [30\ 95\ 7]$$

$$SD \sim [4\ 101\ 33]$$

The principal $\langle 112 \rangle$ pole towards which C is rotating in this case is $R = [1\bar{2}1]$. The projection of this direction in the plane of Fig. 4(a) is given by $R_p = Z \wedge (R \wedge Z) = [24\ 150\ 34]$. The angle between SD and R_p is $\sim 8^\circ$ (within experimental tolerance) confirming that the rotation is towards $[1\bar{2}1]$ and the operating slip system is $(111)[1\bar{2}1]$.

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